

# Winston Workman

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## Objective

Data Science junior specializing in Python-based race analytics and predictive modeling, seeking roles in race strategy, simulation engineering, or sports analytics.

## Education

*The University of North Carolina at Charlotte | Charlotte, NC | Dec 2027 Graduation | BS Data Science*

*Central Piedmont Community College | Charlotte, NC | Graduated July 2025 | Associates in Arts*

### Skills:

- Programming: Python (Pandas, NumPy, Matplotlib, Statsmodels, scikit-learn), R, Java, SQL
- Analysis: Regression, Predictive Modeling, Data Visualization, GeoPandas/GIS, Machine Learning, AI
- Engineering: SolidWorks (CSWA certified), CAD

Relevant Coursework: Databases Design, Data Structures & Modeling, Applied Mechanics, Mechanisms, Electronic Circuits & Devices, Statistics

## Projects

*IMSA GTD Pro Tire Degradation Lap Time Model | Data Science Portfolio Project | 2026*

- Built an end-to-end machine learning pipeline on IMSA GTD Pro race data (tobi/imsa\_data) to model lap time as a function of tire age, cleaning for pit laps, cautions, and traffic-affected laps
- Compared linear regression against gradient boosting using MAE as a percent of lap time, with gradient boosting performing best (0.696% vs. 0.730% baseline,  $R^2 = 0.08$ ) and linear regression barely beating baseline (0.721%)
- Quantified tire degradation at 0.0102% of lap time per lap and documented model limitations and identified fuel load and traffic as next-step features

*NASCAR Stage Caution Pit Strategy Analysis | Independent Data Science Project | 2026*

- Investigated whether pitting during stage-ending caution periods versus staying out affects final race position across 27 NASCAR Cup Series races, using Python and the pynascar API
- Engineered a custom classification variable by cross-referencing pit stop telemetry against caution lap windows to identify strategic pit decisions not directly available in the source data
- Built comparative visualizations in matplotlib revealing that pit strategy effectiveness varies significantly by track type, with pitting favored at short tracks and staying out favored elsewhere

## Experience

*Parts Room Attendant | Riley Technologies | Mooresville, NC | Jun 2024 - Dec 2024*

- Worked within a full-service sports car race team
- Managed national and international shipping/receiving logistics for race car parts and components.
- Conducted parts room inventory audits and tracking to ensure race-critical components were available and accounted for ahead of events
- Supported shop operations including part pickups/deliveries and race trailer preparation, working directly with team equipment and cars

*Operations Internship | Charlotte Motor Speedway | Charlotte, NC | Aug 2023 - Jan 2024*

- Supported track-wide facility operations at a major motorsports venue
- Performed hands-on maintenance and repair work, including tire wall repairs, contributing to the track's readiness for competition and fan safety
- Assisted with parts and equipment logistics across the facility
- Contributed to race-day operations management for the 2023 Roval event, gaining direct experience with the coordination required to execute a major motorsports event

## Additional Experience

*SRO GT World Challenge | CrowdStrike Racing | Virginia International Raceway (VIR) | 2023*

- Engaged directly with drivers, engineers, and team personnel at a professional GT racing event, gaining insight into how a top-tier sports car team operates at the track